

Optimization for Dynamic Laser Inter-Satellite Link Scheduling With Routing: A Multi-Agent Deep Reinforcement Learning Approach

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Abstract—Laser inter-satellite links (LISLs) have greatly extended communication distance between satellites, allowing for establishment of dynamic links to reduce communication delay. However, a closed-loop control is required for LISL, which causes high energy consumption. Proper scheduling of dynamic LISLs can effectively reduce energy consumption and communication delay. In this study, a satellite link mode with three fixed LISLs and one dynamic LISL is designed, and its feasibility is analyzed. The optimization problem is formulated and transformed into a Markov decision process (MDP) by modeling it as a sequential decision problem. By decomposing states, actions, and reward functions, the MDP is divided into the proposed multi-agent deep reinforcement learning (MADRL). Moreover, compressed sensing is utilized to cut down state information to reduce communication, storage, and computation overhead. Furthermore, network parameters and experience sharing, and prioritized experience replay have been adopted to improve stability and convergence speed of network training with a large number of agents. Experimental results show that under different routing strategies, the proposed MADRL can reduce energy consumption by over 15% and delay by approximately two hops compared to fixed LISLs scenario within several iterations.

Index Terms—Laser inter-satellite link, dynamic link, multi-agent deep reinforcement learning, link scheduling, routing strategy.

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I. INTRODUCTION

WIRELESS communication based on ground base stations has been highly successful so far. However, economic cost and construction difficulties make it challenging to cover vast underdeveloped areas, sea areas, deserts, and other remote regions. Moreover, ground base stations are vulnerable to damage in the face of natural disasters such as earthquakes and floods [2]. On the other hand, satellite communication can cover most of the Earth's surface and play a significant role in disaster relief. Nevertheless, the orbit of traditional communication satellites is high, which causes long delays and high propagation loss. Additionally, conventional satellite communications cannot satisfy the needs of a large number of users for high-speed communications simultaneously, and high communication cost reduces the scope of application [3].

The next generation of satellite constellations represented by Starlink enables mega low-Earth orbit (LEO) constellations using technologies such as multiple cubesats launched by one reusable rocket, reducing costs and providing users with more communication resources. Moreover, Starlink plans to adopt laser inter-satellite links (LISLs) in the next generation of constellations. Laser terminals can achieve a communication rate of 10 Gbps at a distance of 4,000 km [4], which significantly enhances overall throughput [5]. The stable establishment of LISLs will eliminate the need for ground relays, improve the throughput, and the propagation delay will be greatly reduced [6], [7], making it different from classic delay-tolerant networks.

However, the transmission distance of LISLs is significant, and the narrow laser beam requires more precise beam alignment. Specifically, space-borne terminals must limit the alignment error to 1 μ rad, and thus a closed-loop feedback mechanism is required to stabilize a link [4]. Therefore, laser terminal must remain active to maintain the link, even during idle periods, which will cause extra energy consumption of the satellite with limited size, weight, and power [8]. Moreover, the fixed four LISLs with adjacent satellites adopted by Starlink underutilizes laser terminals which can realize inter-satellite communications of much longer distance [4], and numerous LISLs are redundant from a network perspective [9]. The feasibility of permanent and temporary links between non-adjacent satellites in Starlink constellation was explored in [10], and the simulation results in [11]

demonstrated that temporary links bypassing the intermediate satellites can effectively reduce communication delay in future satellite communication. Furthermore, as demonstrated in [6] and [12], with the increase of constellations, the routing of satellite network will become increasingly complex. Additionally, the study in [6] shows that, compared to the shortest path, the delay penalty of sub-short path is often not significant. Moreover, pursuing the shortest delay may lead to an unbalanced network load, indicating that the balance between delay and network load requires further research. Furthermore, in other dynamic networks characterized by evolving topologies and time-variable nodes, such as wireless sensor networks [13], Internet of Vehicles [14] and Internet of Things [15], researchers have also optimized routing and link scheduling to improve communication efficiency, yielding promising outcomes. Therefore, the link scheduling and routing strategy of satellite networks are essential topics worth considering.

There has been some progress in the field of satellite network routing and dynamic link scheduling. A method employing reinforcement learning was suggested to optimize the total throughput in a three-layer system [16], where the state was compressed by the compressed sensing (CS) method [17]. Moreover, an energy-efficient route was devised based on a study of satellite battery characteristics [8]. The scheduling of dynamic link scheduling has been explored in [18], [19], [20], [21], [22], and [23], involving the optimization of channel capacity and delay through genetic algorithms in [18] and [19]. Approximate solutions were also presented for optimizing the signal-to-noise ratio of dynamic links and resource allocation for communication [20]. Furthermore, medium-Earth orbit (MEO) and geosynchronous orbit (GEO) satellites were set as relays for LEO satellites, with inter-layer links established to minimize the hop count [21]. Additionally, over polar regions, scheduling prevented frequent link handovers [22], and the allocation of links and power was examined in [23]. Certain studies have also delved into combining routing and link scheduling techniques [24], [25], [26]. The throughput of a delay-tolerant LEO satellite network with radio frequency (RF) links was enhanced by utilizing link scheduling and optimizing routing [24]. Then, a flexible satellite topology scheduling achieved swift network reconstruction and mitigated routing performance decline [25]. Additionally, the scheduling and routing of LISLs were directly solved by mixed integer linear programming in a small constellation [26].

However, traditional methods become significantly more complex when the number of optimization variables increases. Deep reinforcement learning (DRL) can effectively address this challenge and has attracted widespread attention in satellite scenarios [27], [28]. Therefore, applying DRL to satellite routing and resource scheduling is a natural choice. DRL was adopted to achieve routing for a single data flow in [29], while multiple flows were researched in [30]. Moreover, DRL was also employed to implement a dynamic inter-plane link scheduling algorithm based on the signal-to-noise ratio and distance between satellites [31], and to manage the communication capacity [32]. In addition, decomposing the scale of problem is also a feasible solution to reduce complexity.

Therefore, sliding windowing was utilized to reduce the size of linear programming [24], and Benders decomposition was applied to simplify the optimization problem [33].

The above works are effective, however, there are still some shortcomings that need to be addressed. In comparison to terrestrial networks, the periodic patterns of satellite networks in both time and space domains not only simplify the problem, but also enable the design of link modes to reduce energy consumption and average hop count. Moreover, lasers have considerably narrower beam widths than RF signals, resulting in negligible inter-beam interference but requiring more precise alignment, and the number of establishing LISLs via a satellite is strictly bounded by the quantity of space-borne laser terminals. Consequently, the link model for LISLs differs from that of RF links, hence the direct application of RF-based research to LISLs is challenging. Although numerous studies discuss the benefits of dynamic scheduling for optical satellite networks, the majority of research concerning LISL scheduling remains inadequately integrated with routing strategies. In addition, the next generation of satellite networks requires real-time data transmission, which is different from conventional delay-tolerant networks, thus imposing stricter requirements on the time complexity of algorithms. As satellite constellations further expand, both traditional methods and DRL encounter challenges in complexity and convergence. Therefore, we propose a dynamic LISL scheduling algorithm combined with specific route strategy for large constellations, and provide novel contributions to the problems:

- 1) A dynamic LISLs scheduling algorithm combined with a specific routing strategy is proposed. The link mode of a satellite with three fixed LISLs and one dynamic LISL is designed and verified to fully utilize the potential of space-borne laser terminals, and the scheduling of dynamic links is optimized to balance the energy consumption caused by closed-loop control of LISLs and the communication delay. The algorithm is implemented across various scenarios to verify its effectiveness. Moreover, the results indicate that the alternating iteration between routing and scheduling strategy could rapidly reach a steady state.
- 2) A multi-agent deep reinforcement learning (MADRL) method is investigated to solve the scheduling of dynamic LISLs in large-scale constellations. By analyzing the utility of each dynamic LISL to the objective function, the global reward is decomposed into respective rewards for each satellite. Network parameters and experience sharing as well as prioritized experience replay are utilized to enable the MADRL on large constellations.
- 3) The scale of the LISLs state matrix is reduced by CS due to the sparse nature of link state information. The compression of state information not only reduces communication and storage overhead, but the compressed data is also fed directly to a deep neural network (DNN), in contrast to other literatures that necessitate recovery before utilization, and thus leading to a reduction in the scale of DNN and computational complexity.

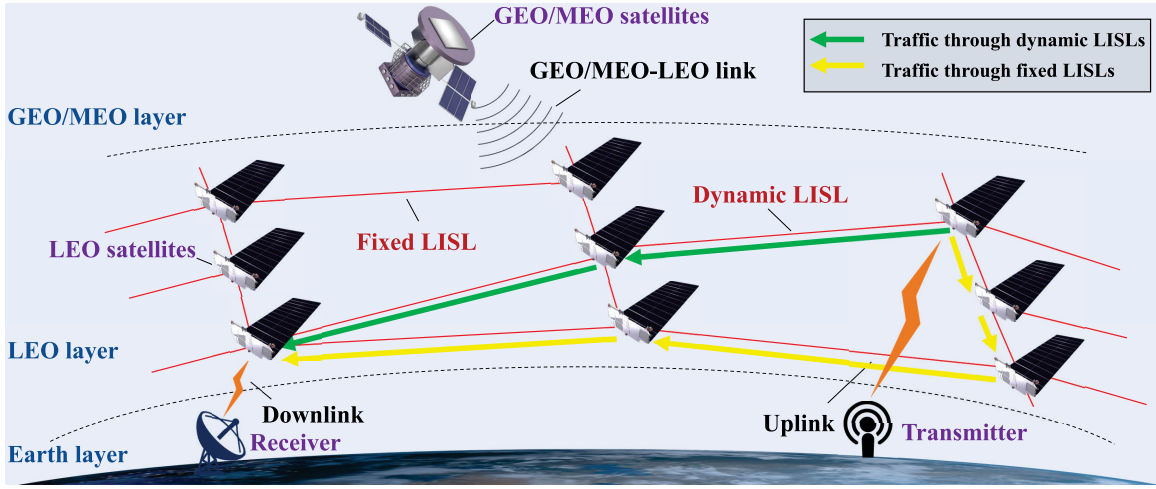


Fig. 1. The system architecture of LEO satellite communication network with dynamic LISLs.

The rest of this paper is organized as follows. Section II introduces the system model, including the architecture of satellite network, LISLs mode, and traffic distribution. Section III introduces the routing strategy, and the optimization problem is formulated. The details of the proposed algorithm are stated in Section IV, including the cooperative MADRL and the structure of the DNN. The results are demonstrated in Section V. Finally, this paper is summarized in Section VI.

II. SYSTEM MODEL

In this section, the system architecture and network model will be introduced. Additionally, the link model of LISLs and data flow distribution will also be described.

A. System Architecture, Constellations and Network Model

As depicted in Fig. 1, the satellite communication network in this paper comprises three layers. The LEO layer is utilized for traffic access and transmission with LISL. The MEO/GEO satellites monitor the status of the LEO layer and relay state information to LEO satellites for resource allocation and the decision of dynamic LISLs scheduling.

Among the three constellations introduced in [5], OneWeb is utilized as the research subject in this paper, which has 18 orbit planes (OPs) at an altitude of 1,200 km and 720 satellites in its constellation. Walker constellation is adopted in OneWeb, where each OP shares the same inclination and the right ascension of ascending node of each plane is evenly distributed in the equatorial plane. In order to avoid collisions among satellites in different orbits, satellites in different orbital planes need to maintain a certain phase difference. The constellation adopted in this paper can be expressed by $i : N/P/F$ notation of the Walker constellation as $85^\circ : 720/18/1$, where i is the orbital inclination, and N , P , and F denote the total number of satellites, the number of orbital planes, and the phase factor in the constellation, respectively. The phase difference $\Delta\phi$ for equivalent satellites in neighboring planes is given by

$$\Delta\phi = \frac{360^\circ}{N}F, \quad (1)$$

resulting in a 0.5° change between satellites in adjacent OPs. The sub-satellite point trajectory is shown in Fig. 3, in which the arrows indicate the direction of satellites' motion.

The satellite network topology can be represented by the directed graph $G = \langle \mathcal{V}, \mathcal{E} \rangle$, where $\mathcal{V}$ is the set of vertices, namely the satellites, and is defined as $\{i \in \mathcal{V} | i = 1, 2, \dots, N\}$. Meanwhile, $\mathcal{E}$ is the set of edges, and $(i, j) \in \mathcal{E}$ if and only if a link from $i \in \mathcal{V}$ to $j \in \mathcal{V}$ is established. Moreover, each link $(i, j) \in \mathcal{E}$ is assigned a capacity $c(i, j)$.

B. Link Model and Classification of LISLs

LISLs exhibit extremely narrow beam widths, which require high accuracy for acquisition, pointing, and tracking. Space-borne laser terminals are utilized to meet these demands. Each laser terminal can establish only one link with another terminal, and since a closed-loop control is necessary to maintain a LISL, if $(i, j) \in \mathcal{E}$, and then $(j, i) \in \mathcal{E}$. Moreover, the number of LISLs that each satellite can establish is strictly limited by the amount of space-borne laser terminals N_{LISL} . The set comprised of satellites with a direct link to satellite k is defined as $\mathcal{N}(k)$, which can be expressed as $\{m \in \mathcal{N}(k) | m \in \mathcal{V}, (m, k) \in \mathcal{E}\}$, and thus $|\mathcal{N}(k)| \leq N_{LISL}$.

The link modes in the same constellation are divided into intra-OP LISLs which are established between two satellites in the same orbit, and inter-OP LISLs which are between satellites in different OPs [10]. Satellites in the same orbit are relatively stationary, and so the intra-OP link is easy to establish. The inter-OP LISL are generally formed between satellites in the same direction. For Starlink system, $N_{LISL} = 4$, where two terminals are adopted for intra-OP links and the others are for connecting satellites on the two sides of the adjacent orbits in the same direction, as shown in Fig. 2(a).

The establishment of temporary links under Starlink constellation was discussed in [11]. Intuitively, this helps to reduce the hop count for message propagation by bypassing intermediate satellites, thus reducing the processing delay of information. Meanwhile, as previously mentioned, due to the uneven distribution of network load, links with low loads can

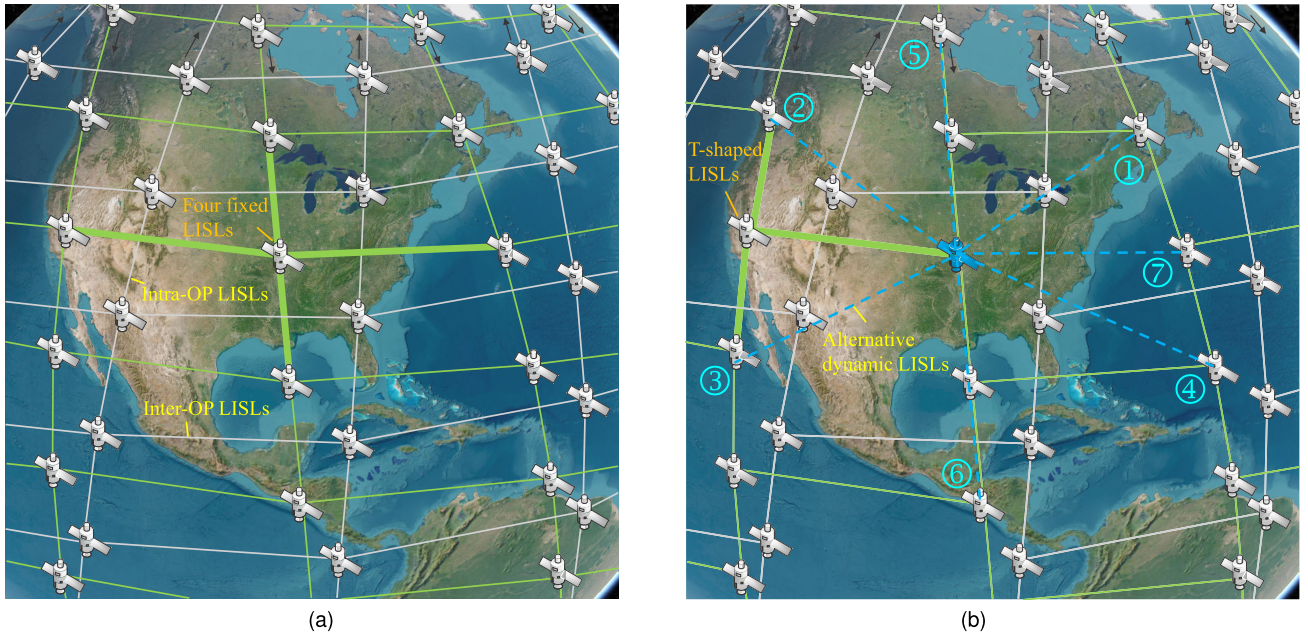


Fig. 2. The diagram of LISLs mode in the constellation. (a) The fixed four LISLs with adjacent satellites mode. (b) The three fixed LISLs and one dynamic LISLs mode, where the fixed LISLs form a T-shaped structure, and the dynamic LISLs have seven alternative linked objects or can be inactive.

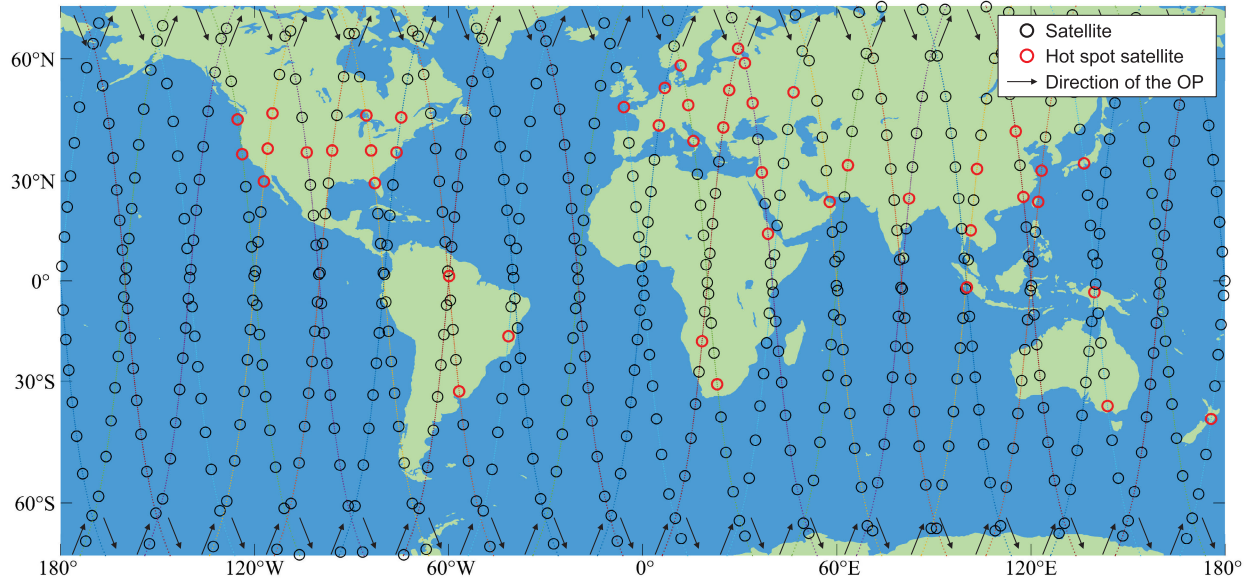


Fig. 3. The sub-satellite point trajectory of the constellation and the hot spot satellites selected at a certain time.

be inactive to reduce the overall power consumption of the whole system. In this paper, to ensure the connectivity of constellation and the basic communication performance, it is guaranteed that each satellite has three fixed links to connect with neighboring satellites, forming a T-shaped structure as shown in Fig. 2(b). Meanwhile, a dynamic LISL can be established between one of the seven alternative satellites or be inactive. The numbers in the figure correspond to the possible options, and zero represents an inactive LISL. Moreover, the topology of the two adjacent satellites shows mirror symmetry, and so the label of the adjacent satellite is also symmetry.

Based on the above discussion, $\mathcal{E}$ can be divided into the fixed LISLs $\mathcal{E}_f$ and the dynamic LISLs $\mathcal{E}_d$, and the feasible region of $\mathcal{E}_d$ is denoted as Ω . Since V and $\mathcal{E}_f$ are constant, there is a one-one mapping between G and $\mathcal{E}_d \subset \Omega$, and the

G can be rewritten as $G(\mathcal{E}_d)$. Additionally, the requirements to establish dynamic LISLs are analyzed in Appendix-A, as well as the interference between dynamic and fixed LISLs in Appendix-B. The results indicate all possible dynamic LISLs are achievable with the terminals introduced by [4]. Moreover, interference between LISLs is negligible because of the extremely narrow laser beams. Consequently, satellites can simultaneously establish multiple LISLs for communications.

C. Traffic Distribution

Due to differences in economic development and population density distribution, the majority of communication demands are concentrated in a small number of highly developed land areas. Consequently, only a small number of satellites in a

constellation will carry most of the traffic at a time [8]. In this paper, these satellites are referred to as “hot spot” satellites. Specifically, satellites primarily located in North America, Europe, and Asia-Pacific regions are selected as hot spot satellites set $\mathcal{V}^h \subset \mathcal{V}$, and are highlighted in red in Fig. 3. An “episode” is defined as the temporal interval between two consecutive handovers of the traffic access point, triggered by the high-speed motion of the satellite. K satellites are randomly chosen from $\mathcal{V}^h$ as the source and destination nodes set $\mathcal{V}_0$ and remain fixed for a given episode. Traffic is transmitted between each node in the $\mathcal{V}_0$, and so there are $K(K-1)$ data flows. Due to the motion of satellites and changes in traffic demands, $\mathcal{V}_0$ will be reselected from $\mathcal{V}^h$ in the next episode.

Due to the rapid motion of LEO satellites, each episode lasts merely a few minutes. The statistical attributes of data flow exhibit minor alteration at this scale. Consequently, the distribution of data flow volume is assumed to be stationary within one episode. An episode is divided into T time slots, and the traffic is evenly distributed in each time slot, while the volume of data flow in time slot t from $i \in \mathcal{V}_0$ to $j \in \mathcal{V}_0$ is written as f_{ij}^t , and its transmission should be completed within the current slot. Moreover, f_{ij}^t is composed of independent requests as

$$f_{ij}^t = a_{ij}^1(t) + a_{ij}^2(t) + \dots + a_{ij}^n(t), \quad t = 0, 1, \dots, T-1, \quad (2)$$

where n is the total number of users in the service area of the i -th satellite during time slot t , and $a_{ij}^k(t)$ represents the size of the message sent by the user k to f_{ij}^t in time slot t . Assume that for different k , $a_{ij}^k(t)$ are independent and identically distributed. Considering the rapidly growing demand for LEO satellite networks, exemplified by Starlink serving 1.5 million users, it can be inferred that n will reach several hundred. Therefore, f_{ij}^t approximately follows the Gaussian distribution, according to the central limit theorem, similar to the diffusion approximation in queue theory [34]. The data flow matrix $\mathbf{F}^t$ in time slot t is given by

$$\mathbf{F}^t = \begin{pmatrix} 0 & f_{v_1 v_2}^t & \dots & f_{v_1 v_K}^t \\ f_{v_2 v_1}^t & 0 & \dots & f_{v_2 v_K}^t \\ \vdots & \vdots & \ddots & \vdots \\ f_{v_K v_1}^t & f_{v_K v_2}^t & \dots & 0 \end{pmatrix}, \quad (3)$$

where v_k refers to the k -th satellite in the $\mathcal{V}_0$. And the total throughput in time slot t can be represented as $\|\mathbf{F}^t\|_{m1}$, where $\|\cdot\|_{m1}$ means the sum of the absolute value of all elements in the matrix. The $\mathbf{F}^t$ of each slot follows the same distribution during one episode.

III. PROBLEM FORMULATION

The routing strategy and the optimization problem for LISLs scheduling are discussed in this section.

A. Routing Strategy of Data Flow

For traffic routing, when the source and destination nodes set $\mathcal{V}_0$ and the corresponding data flows f_{ij}^t are given, the only

thing that needs to be determined is the traffic distribution of the flows over edges. The ratio of the flow f_{ij}^t on edge $(u, v) \in \mathcal{E}$ is denoted as $x_{ij}(u, v)$. Moreover, the following constraints should be satisfied.

- 1) The link capacity cannot be exceeded, which can be expressed as

$$\sum_{i,j \in \mathcal{V}_0, i \neq j} f_{ij}^t \cdot x_{ij}^t(u, v) \leq c(u, v), \quad \forall (u, v) \in \mathcal{E}. \quad (4)$$

- 2) For f_{ij}^t , when it flows through an intermediate node k , the flow entering and leaving the node should be equal, which can be described as

$$\sum_{m \in \mathcal{N}(k)} (x_{ij}^t(k, m) - x_{ij}^t(m, k)) = 0, \quad \forall i, j \in \mathcal{V}_0, i \neq j, \text{ when } k \neq i, k \neq j, \quad (5)$$

- 3) For the source node of f_{ij}^t , the total data flow must exit, which can be written as

$$\sum_{m \in \mathcal{N}(i)} (x_{ij}^t(i, m) - x_{ij}^t(m, i)) = 1, \quad \forall i, j \in \mathcal{V}_0, i \neq j. \quad (6)$$

- 4) The data flow f_{ij}^t should totally enter the satellite j as

$$\sum_{m \in \mathcal{N}(j)} (x_{ij}^t(j, m) - x_{ij}^t(m, j)) = -1, \quad \forall i, j \in \mathcal{V}_0, i \neq j. \quad (7)$$

However, a routing loop is still possible when the above constraints are met, for instance, a routing loop without a source can satisfy all the above constraints, but it is not in line with the actual situation. In [35], an algorithm constructing a directed acyclic graph was proposed, while a simpler treatment is adopted in this paper by adding an additional term as

$$H_{avg} = \frac{\sum_{(u,v) \in \mathcal{E}} \sum_{i,j \in \mathcal{V}_0, i \neq j} x_{ij}^t(u, v) \cdot f_{ij}^t}{\|\mathbf{F}^t\|_{m1}}, \quad (8)$$

which means the average hop count of the data flows in satellite network. If the goal is to minimize the target function g , a large coefficient $\mathcal{L}$ can multiply g , and then add with H_{avg} . As the simulation results in [11], the difference of node delay plays an important role in different paths. Therefore, in this paper, the average hop count is reduced to optimize the delay, and three routing policies are designed as follows.

- 1) $\mathcal{R}_1$: Minimize the average hop count. Eq. (8) is selected as the objective function and the aforementioned constraints (4)-(7) are applied, the problem can be written as

$$\min_{x_{ij}^t(u,v)} \left(\frac{\sum_{(u,v) \in \mathcal{E}} \sum_{i,j \in \mathcal{V}_0, i \neq j} x_{ij}^t(u, v) \cdot f_{ij}^t}{\|\mathbf{F}^t\|_{m1}} \right), \quad \text{s.t. (4), (5), (6), (7)}. \quad (9)$$

- 2) $\mathcal{R}_2$: Minimize the link load. Let z denote the maximum load of link, which is regarded as the optimization objective. Another constraint needs to be added to ensure that the load rate of each link does not exceed z . Moreover, the routing loop should be avoided by adding

the term defined in (8). Therefore, this routing strategy can be calculated by

$$\begin{aligned} \min_{x_{ij}^t(u,v)} & \left(\mathcal{L} \cdot z + \frac{\sum_{(u,v) \in \mathcal{E}} \sum_{i,j \in \mathcal{V}_0, i \neq j} x_{ij}^t(u,v) \cdot f_{ij}^t}{\|\mathbf{F}^t\|_{m1}} \right), \\ \text{s.t.} & \text{ (4), (5), (6), (7),} \\ & \frac{\sum_{i,j \in \mathcal{V}_0, i \neq j} f_{ij}^t \cdot x_{ij}^t(u,v)}{c(u,v)} \leq z, \quad \forall (u,v) \in \mathcal{E}, \end{aligned} \quad (10)$$

where $\mathcal{L}$ is a large coefficient as mentioned before.

- 3) $\mathcal{R}_3$: Compromise between average hop count and link load. Similar to the previous discussion, the problem can be expressed as

$$\begin{aligned} \min_{x_{ij}^t(u,v)} & \left(z + \frac{\sum_{(u,v) \in \mathcal{E}} \sum_{i,j \in \mathcal{V}_0, i \neq j} x_{ij}^t(u,v) \cdot f_{ij}^t}{\|\mathbf{F}^t\|_{m1}} \right), \\ \text{s.t.} & \text{ (4), (5), (6), (7),} \\ & \frac{\sum_{i,j \in \mathcal{V}_0, i \neq j} f_{ij}^t \cdot x_{ij}^t(u,v)}{c(u,v)} \leq z, \quad \forall (u,v) \in \mathcal{E}. \end{aligned} \quad (11)$$

These constraints and objective functions are all linear, and so the traffic distribution $x_{ij}^t(u,v)$ can be obtained easily by the linear programming tool.

B. Optimization Problem Formulation

When the routing policy $\mathcal{R}$ is determined, due to the $\mathcal{V}_0$ is fixed during the current episode, the traffic distribution only depends on the network topology $G(\mathcal{E}_d)$ and the volume of data flow $\mathbf{F}^t$, and so the $x_{ij}^t(u,v)$ can be written as $x_{ij}^t(u,v, G(\mathcal{E}_d), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R})$. Since $\mathbf{F}^t$ is a random variable, a reasonable $G(\mathcal{E}_d)$ is expected to be optimized to reduce the average hop count and energy consumption. The average hop count in this case can be written as

$$\begin{aligned} H(G(\mathcal{E}_d), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R}) \\ = \frac{\sum_{(u,v) \in \mathcal{E}} \sum_{i,j \in \mathcal{V}_0, i \neq j} x_{ij}^t(u,v, G(\mathcal{E}_d), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R}) \cdot f_{ij}^t}{\|\mathbf{F}^t\|_{m1}}. \end{aligned} \quad (12)$$

Assuming that $\mathbf{F}^t$ is composed of a large amount of data streams, and so $\|\mathbf{F}^t\|_{m1}$ does not fluctuate much. The term related to average hop count in the objective function is rewritten as

$$\begin{aligned} H_0(G(\mathcal{E}_d), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R}) \\ = \sum_{(u,v) \in \mathcal{E}} \sum_{i,j \in \mathcal{V}_0, i \neq j} x_{ij}^t(u,v, G(\mathcal{E}_d), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R}) \cdot f_{ij}^t. \end{aligned} \quad (13)$$

Meanwhile, energy consumption is directly related to the number of LISLs, and the energy consumption term is set as

$$P_0(G(\mathcal{E}_d)) = |\mathcal{E}_d|, \quad (14)$$

since the number of fixed LISLs is determined, it is not counted in it. And the problem can be formulated as

$$\max_{\mathcal{E}_d \subset \Omega} \left(-\beta \mathbb{E}_{\mathbf{F}^t} [H_0(G(\mathcal{E}_d), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R})] - (1 - \beta) P_0(G(\mathcal{E}_d)) \right), \quad (15)$$

where β is used to achieve a trade-off between two goals. When $\beta = 1$, the objective function only focuses on the average hop count, and when $\beta = 0$, only the energy consumption is optimized.

IV. MADRL ALGORITHM FOR DYNAMIC LISLs SCHEDULING

In this section, the optimization defined in (15) is translated into a Markov decision process (MDP). Moreover, the MDP is decomposed into multi-agent reinforcement learning (MARL) to reduce the complexity, while the deep Q-learning network (DQN) [36] is adopted to solve the problem. Meanwhile, based on the characteristics of satellite networks, several methods are also introduced to reduce communication, storage, and computation overhead and accelerate convergence.

A. Cooperative MARL for Dynamic Link Scheduling

It is challenging to solve (15) directly, as the feasible region is discrete and extremely large. However, since $\mathcal{V}_0$ is constant and $\mathbf{F}^t$ follows the same distribution within one episode, the link status in a time slot can provide a reference for the selection of $\mathcal{E}_d$ in subsequent time slots. Therefore, as an alternative, in the time slot 0 of the current episode, the satellite network topology remains unchanged from the last time slot $T - 1$ of the previous episode, and for each time slot t , the next time slot dynamic LISLs $\mathcal{E}_d^{t+1}$ are optimized according to the current link state $\mathbf{S}^t$ as

$$\mathcal{E}_d^{t+1} = \pi(\mathbf{S}^t), \quad t = 0, 1, \dots, T - 2, \quad (16)$$

where $\pi(\cdot)$ is the policy that adjusts $\mathcal{E}_d^{t+1}$ according to $\mathbf{S}^t$, and $\mathcal{E}_d^{t+1}$ is added a superscript to represent the dynamic LISLs in time slot $t + 1$, which is the same for $\mathcal{E}^{t+1}$. Therefore, formula (15) can be reorganized as

$$\begin{aligned} \max_{\pi} & \left(-\beta \mathbb{E}_{\mathbf{F}^{t+1}} [H_0(G(\pi(\mathbf{S}^t)), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R})] \right. \\ & \left. - (1 - \beta) P_0(G(\pi(\mathbf{S}^t))) \right), \quad t = 0, 1, \dots, T - 2. \end{aligned} \quad (17)$$

Under this strategy, the scheduling of $\mathcal{E}_d^{t+1}$ in one episode is modeled as a sequential decision, and the link state of the current time slot only depends on the last one, thus forming an MDP. The MDP can be described with a quintuple $\langle S, A, \mathbf{P}, R, \gamma \rangle$, where S , A , and $\mathbf{P}$ represent the set of the environment state, the action space, and the state transition probability matrix, respectively, and $\gamma \in [0, 1]$ is the discount factor. Concretely, in dynamic link scheduling problem, these variables have the following meanings.

- 1) **State**: The topology of the satellite network and the load of the LISLs in time slot t can be regarded as the environment state, and can be defined as

$$\mathbf{S}^t = \begin{pmatrix} s_{11}^t & s_{12}^t & \cdots & s_{1N}^t \\ s_{21}^t & s_{22}^t & \cdots & s_{2N}^t \\ \vdots & \vdots & \ddots & \vdots \\ s_{N1}^t & s_{N2}^t & \cdots & s_{NN}^t \end{pmatrix}, \quad (18)$$

where N is the total number of the satellites, and s_{uv}^t can be obtained as

$$s_{uv}^t = \begin{cases} \sum_{i,j \in \mathcal{V}_0, i \neq j} f_{ij}^t x_{ij}^t(u, v, G(\mathcal{E}_d^t), \mathbf{F}^t | \mathcal{V}_0, \mathcal{R}), & \text{if } (u, v) \in \mathcal{E}^t, \\ 0, & \text{others.} \end{cases} \quad (19)$$

- 2) **Action**: Considering the whole satellite network as an agent, the action it takes in the time slot t is the change to the $\mathcal{E}_d$ in the next time slot, therefore the action in time slot t is given by

$$A^t = \mathcal{E}_d^{t+1} = \pi(\mathbf{S}^t), \quad t = 0, 1, \dots, T-2, \quad (20)$$

in which A^t means the action it adopts in time slot t .

- 3) **Reward**: The reward function represents the benefits gained by executing action A^t in the state $\mathbf{S}^t$. According to the optimization target in (17), the reward can be calculated as

$$R^t(\mathbf{S}^t, A^t) = -\beta(H_0(G(\pi(\mathbf{S}^t)), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R})) - (1 - \beta)P_0(G(\pi(\mathbf{S}^t))), \quad t = 0, 1, \dots, T-2, \quad (21)$$

where R^t denotes the achieved reward after the action A^t in the state $\mathbf{S}^t$. The first term represents the improvement of average hop count in time slot $t+1$, and the second one is the energy consumption.

However, the MDP described in (17) to (21) has an infinite state space. In addition, given that $A^t = \mathcal{E}_d^{t+1} \subset \Omega$, the action space is also extensive, making it challenging to optimize dynamic LISLs of the satellite network as a whole. To address this issue, a practical approach is to reduce the action space by decomposing the problem into independent decisions made by each satellite concerning its own dynamic link selection. In this context, each satellite in the constellation acts as an agent and makes decisions based on its own state information. By considering the decisions of all satellites, the dynamic links $\mathcal{E}_d^{t+1}$ in the next time slot are ultimately determined, thus a cooperative MARL is formed. Therefore, for each satellite, environment state S , action A , and reward function R are adjusted correspondingly as follows.

- 1) **State for satellite i** : For $i \in \mathcal{V}$ in time slot t , select the satellites in the range of $(2a+1) \times (2b+1)$ centered on satellite i to form a set $\mathcal{V}_{near}^i$. The state for satellite i is defined as

$$\mathbf{s}_i^t = \begin{pmatrix} s_{i_1 i_1}^t & s_{i_1 i_2}^t & \cdots & s_{i_1 i_L}^t \\ s_{i_2 i_1}^t & s_{i_2 i_2}^t & \cdots & s_{i_2 i_L}^t \\ \vdots & \vdots & \ddots & \vdots \\ s_{i_L i_1}^t & s_{i_L i_2}^t & \cdots & s_{i_L i_L}^t \end{pmatrix}, \quad (22)$$

where the subscript i_k denotes the k -th satellite in $\mathcal{V}_{near}^i$, and s_{mn}^t is defined in (19). It is obvious that $\mathbf{s}_i^t$ is a submatrix of $\mathbf{S}^t$. Moreover, as illustrated in Section II-C, the topology of the two adjacent satellites shows mirror symmetry, for the mirrored satellites, their state matrices are flipped left and right after calculating the results in (22).

- 2) **Action for satellite i** : The satellite i can take the action a_i^t to determine the dynamic LISLs type as

$$a_i^t = 0, 1, \dots, a_{max}, \quad (23)$$

where $a_i^t = 0$ means the satellite determines to make dynamic LISL inactive, while other options corresponding to each number have been introduced in Section II-C, thus $a_{max} = 7$, and similarly, for the mirrored satellites, these options are also flipped accordingly. Meanwhile, the dynamic link will be established in the next time slot only if the satellites at both ends of the link make the decision to establish it simultaneously.

- 3) **Reward for satellite i** : The total objective function is adopted as the reward function for each agent in [27] and [31], which is a convenient method. However, in these works, the number of agents is not large and a single agent has a large impact on the total reward, which is not applicable in the case of this paper. With hundreds of satellites in the constellation, a single dynamic link has little effect on the overall traffic distribution. Consequently, it may become challenging for the agent to learn from the rewards of taking actions. Therefore, it is necessary to split the rewards so that each agent can intuitively evaluate the benefits of its actions. Since adding a constant to the objective function does not affect the result, Eq. (17) can be rewritten as

$$\begin{aligned} \max_{\pi} & \left(-\beta \mathbb{E}_{\mathbf{F}^{t+1}} [H_0(G(\pi(\mathbf{S}^t)), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R}) \right. \\ & \quad \left. - H_0(G(\mathcal{E}_d^{t+1}), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R})) \right. \\ & \quad \left. - (1 - \beta)P_0(G(\pi(\mathbf{S}^t))) \right), \quad t = 0, 1, \dots, T-2, \end{aligned} \quad (24)$$

where the added $H_0(G(\mathcal{E}_d^{t+1}), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R})$ denotes the total amount of the load of the LISLs when there is no established dynamic link. Correspondingly, the global reward in (21) can be reorganized as

$$\begin{aligned} R^t(\mathbf{S}^t, A^t) &= -\beta(H_0(G(\pi(\mathbf{S}^t)), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R}) \\ & \quad - H_0(G(\mathcal{E}_d^{t+1}), \mathbf{F}^{t+1} | \mathcal{V}_0, \mathcal{R})) \\ & \quad - (1 - \beta)P_0(G(\pi(\mathbf{S}^t))), \quad t = 0, 1, \dots, T-2. \end{aligned} \quad (25)$$

Therefore, the impact of the action adopted by each satellite on global reward in (25) should be investigated. If a satellite does not establish a dynamic link, it will have no impact on global reward. Conversely, if a dynamic link is formed, the energy consumption term in the reward function will increase by 2, since this LISL will be counted twice in $\mathcal{E}_d^{t+1}$ for different directions. Without considering the influence of other dynamic LISLs, if a dynamic LISL is established, the average hop count term in the global reward function will decrease. Specifically, if a dynamic LISL in modes 1 to 6 is established, this term will decrease by the same value as the traffic passing through this dynamic link, since the traffic will take an additional hop to reach the same destination through the fixed LISLs. For the traffic passing through the dynamic LISL in mode 7, part of

the traffic can be equivalently transmitted through other fixed LISLs without increasing the number of hops, and the other part of the traffic needs two more hops, thus the traffic through the LISL in mode 7 is multiplied by a factor η . Ultimately, the reward function of satellite i is expressed as

$$r_i^t = \begin{cases} 0.5 (\beta \cdot (s_{ij}^{t+1} + s_{ji}^{t+1}) - 2(1 - \beta)), & \text{if } j \in \mathcal{N}(i), (i, j) \in \pi(\mathbf{S}^t), a_i^t = 1, \dots, 6, \\ 0.5 (\beta \cdot \eta(s_{ij}^{t+1} + s_{ji}^{t+1}) - 2(1 - \beta)), & \text{if } j \in \mathcal{N}(i), (i, j) \in \pi(\mathbf{S}^t), a_i^t = 7, \\ 0, & \text{others,} \end{cases} \quad (26)$$

where s_{ij}^{t+1} is defined in (19), and each term is multiplied by 0.5 because a dynamic LISL is counted twice when the reward is calculated separately for each satellite. Since the influence of other dynamic LISLs is not considered, the reward function calculated in (26) tends to be overestimated when there are a large number of dynamic links. However, results in the subsequent section will indicate that there is a strong linear relationship between the sum of the reward functions obtained by (26) for each satellite and global reward in (25), which proves that the decomposition method is effective.

B. Deep Q-Learning Network

For the MARL constructed in the previous subsection, which operates in a continuous state space with a limited action space, it is appropriate to utilize a DQN to effectively learn the optimal scheduling policy for the dynamic LISLs. Each agent, namely a satellite, obtains Q-value function $Q_i(s_i^t, a_i^t; \theta)$, which describes the expected reward of taking a particular action in a particular state, by the DNN with the parameters θ . To balance the exploration and exploitation in DRL, an ϵ -greedy strategy is adopted to determine action a_i^t chosen by the agent i as

$$a_i^t = \begin{cases} \arg \max_a Q(s_i^t, a; \theta), & \text{with probability } 1 - \epsilon, \\ \text{random action,} & \text{with probability } \epsilon, \end{cases} \quad (27)$$

where ϵ denotes the probability of an exploration by the agent. At the beginning of training, the value of ϵ is set close to 1 to encourage the agent to explore more, while when the training is stabilized, ϵ tends to approach 0 to enable the agent to concentrate on optimizing the existing policy. After the agent takes action, it goes to the next state s_i^{t+1} and gets a reward r_i^t , then the resulting experience sample $\langle s_i^t, a_i^t, r_i^t, s_i^{t+1} \rangle$ are stored in experience pool, and the latest sample will cover the earliest one, if the given volume of experience pool is exceeded.

The network utilizes the sample stored in experience pool to update the parameters according to Bellman equation, which is given by

$$Q_i(s_i^t, a_i^t; \theta) = r_i^t + \gamma \max_{a_i^{t+1}} Q_i(s_i^{t+1}, a_i^{t+1}; \theta). \quad (28)$$

In order to avoid an overestimation of Q-value function, Double DQN [37] adopts two DNNs with the same structure,

which are the current network $Q_i(s_i^t, a_i^t; \theta)$ and the target network $Q_i(s_i^t, a_i^t; \theta^-)$. In the training period, according to the sample $\langle s_i^t, a_i^t, r_i^t, s_i^{t+1} \rangle$ selected from experience pool, the current value function is calculated as

$$\begin{aligned} \widehat{Q}(s_i^t, a_i^t) \\ = r_i^t + \gamma Q_i(s_i^{t+1}, \arg \max_a Q_i(s_i^{t+1}, a; \theta); \theta^-). \end{aligned} \quad (29)$$

Then, the MSE-based loss function L_{MSE} is expressed as

$$L_{\text{MSE}} = (\widehat{Q}(s_i^t, a_i^t) - Q(s_i^t, a_i^t; \theta))^2, \quad (30)$$

and if the input is in the form of a minibatch, the loss function is averaged according to the batch size. The parameters of the current network θ are updated by the adaptive moment estimation method [38] according to the loss function and those of the target network θ^- are replaced by θ in each C slots.

The reward obtained by the agent is strongly related to its current state. When the current surrounding load is heavy, establishing a dynamic link can generally obtain more rewards and vice versa. Based on this, the Dueling DQN [39] is adopted which splits the output into two streams, one for estimating the state value function $V(s)$, and the other one for estimating the advantage function $A(s, a)$ of each action a . As a result, the Q-value function is obtained by

$$Q(s, a) = V(s) + A(s, a) - \bar{A}(s, a), \quad (31)$$

where $\bar{A}(s, a)$ denotes the average over different a .

C. Framework of the Algorithm

The framework of the MADRL is illustrated in Fig. 4. In order to account for the limited satellite resources and ensure the stability of the training with a large number of agents in the system, the following techniques have been adopted.

1) *Compress Sensing for State Information*: The state information s_i^t is transmitted frequently and stored in large quantities in the experience pool, and this causes a lot of communication and storage overhead. Therefore, compressing state information is of great value to reduce the requirements of the transmission bandwidth and storage capacity, as well as to cut down the network size.

Since the number of LISLs that each satellite established is not more than $N_{\text{LISL}} = 4$, the amount of nonzero elements in each column of s_i^t is significantly smaller than the length L of each column. Therefore, the compressed state $\tilde{s}_i^t$ can be obtained by

$$\tilde{s}_i^t = \Phi_0 s_i^t, \quad (32)$$

where Φ_0 is a fixed $D \times L$ matrix randomly generated from the Gaussian distribution, and so the state information is compressed from $L \times L$ to $D \times L$. Furthermore, as CS results in low information loss and the position of the non-zero elements in s_i^t is known, it is feasible to adopt the $\tilde{s}_i^t$ directly as input to the neural network with low performance degradation, which reduces the network parameters. In addition, due to the weak coherence of Φ_0 , the input is equivalent to being normalized, which also speeds up the convergence of the network.

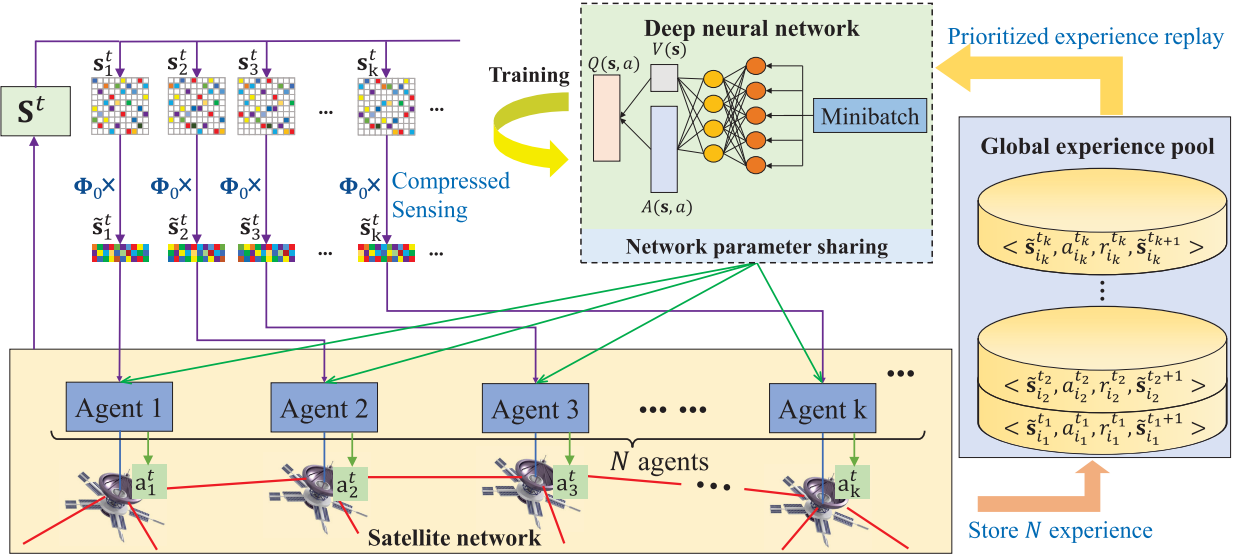


Fig. 4. The architecture of the MADRL for dynamic LISLs scheduling.

TABLE I
SIMULATION PARAMETERS

PARAMETERS	VALUES
SATELLITE NETWORK PARAMETERS	
Satellite altitude	1200 km
Number of satellites N	720
Number of OPs P	18
Inclination of OPs ϕ	85
Phase factor of constellation F	1
Number of satellites in $\mathcal{V}^h$	45
Number of satellites in $\mathcal{V}_0$ K	15
Number of the data streams	210
The mean rate of the data streams f_{ij}	1.3 Gbps
The standard deviation of the data streams f_{ij}	0.2 Gbps
The link capacity between u and v $c(u, v)$	10 Gbps
The range of the state for satellite i	9×9
The size of sensing matrix Φ_0	10×81
The reward factor for mode 7 η	0.95
DRL PARAMETERS	
Training episode	1500
Time slots every episode T	10
Learning rate	10^{-4}
Start learning size N_s	4×10^5
Maximum size of experience pool	10^6
The discount factor γ	0.95
Current network learning frequency	10
Target network update epochs C	200
Minibatch size	2048
Exploration probability ϵ	[0,1]

2) *Network Parameter and Experience Sharing*: In cooperative MADRL, the environment encountered by a single agent is non-stationary due to the changing strategies of other agents, which poses a challenge to the stability of network training [40]. To ensure stability and speed up convergence, the

satellite network of homogeneous agents can utilize the same network parameters, which guarantees that all agents follow the same policy. Additionally, storing experience samples of all agents in a global experience pool and sharing them during training can improve the generalization ability and robustness of the network, and also reduce the training overhead as well. The implementation of network parameter and experience sharing is shown in Fig. 4.

3) *Prioritized Experience Replay*: Once the number of samples in experience pool N_e exceeds a threshold N_s , samples are drawn from experience pool to train the DNN. However, a dynamic LISL is only established when both agents agree to establish the link, which results in sparse rewards for the agent at the start of training. Consequently, the experience pool contains only a few experiences that establish dynamic LISLs successfully, making it challenging for the network to efficiently learn useful information. To address this issue, the prioritized experience replay method [41] is employed to sample in experience pool during the initial training stage. Different from the original method in [41], the sampling probability p_i of a sample i is determined by the absolute value of its reward function as

$$p_i = \begin{cases} \frac{|r_i|^\alpha}{\sum_{j=1}^{N_e} |r_j|^\alpha}, & \text{if episode } k < k_{th}, \\ \frac{1}{N_e}, & \text{others,} \end{cases} \quad (33)$$

where r_i indicates the reward of sample i and α adjusts the priority, making it easier to sample successful LISL establishment experiences from experience pool in the initial stage, and k and k_{th} represent the current episode and the threshold episode, respectively, which means that each sample is collected with equal probability when the training stabilizes. This method accelerates the learning of the network by giving priority to samples with higher absolute values of reward function.

Finally, the detailed description of the MADRL algorithm for scheduling of dynamic LISLs is shown in **Algorithm 1**.

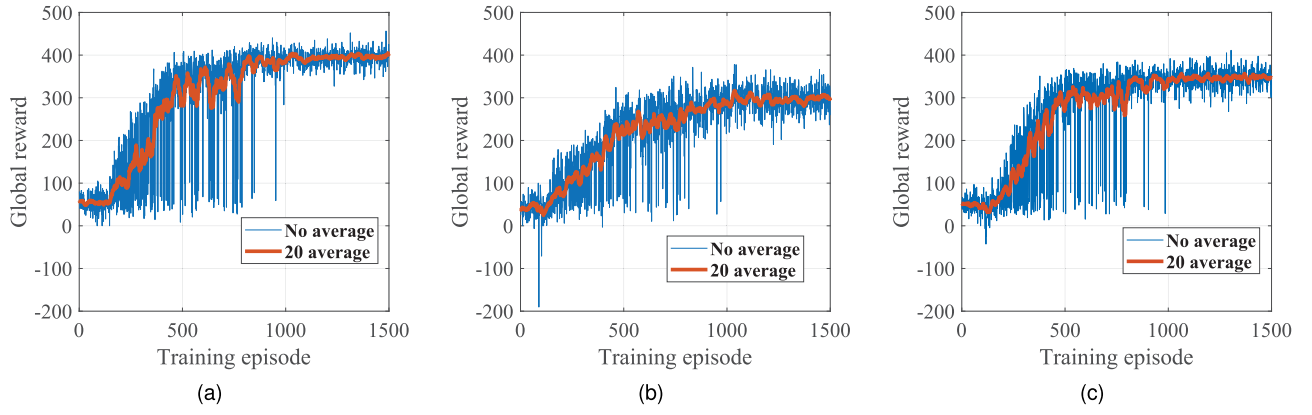


Fig. 5. The episode reward with $\beta = 0.5$ under different routing strategies. (a) Strategy $\mathcal{R}_1$. (b) Strategy $\mathcal{R}_2$. (c) Strategy $\mathcal{R}_3$.

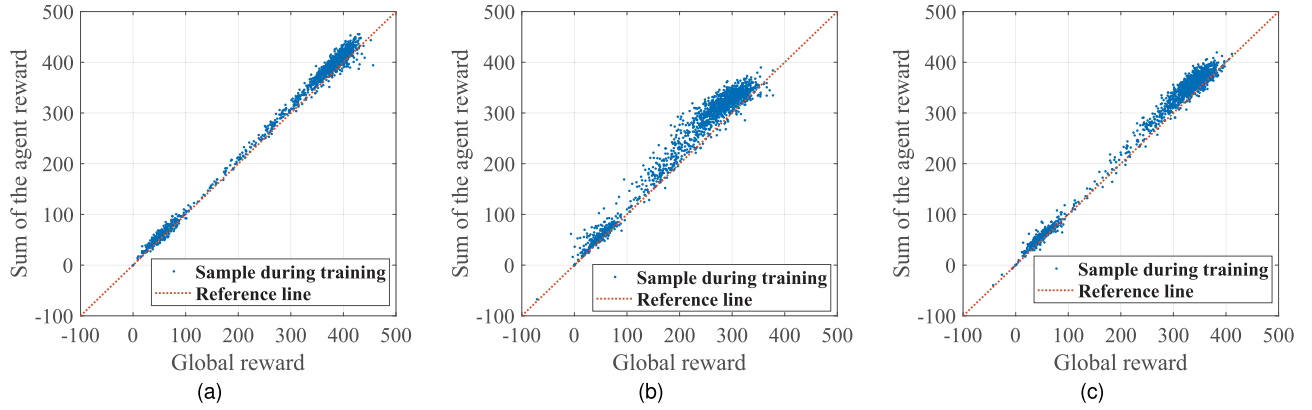


Fig. 6. The global reward and the sum of agent rewards, $\beta = 0.5$. (a) Strategy $\mathcal{R}_1$. (b) Strategy $\mathcal{R}_2$. (c) Strategy $\mathcal{R}_3$.

V. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

A. Simulation Parameters

The update of exploration probability ϵ is expressed as

$$\epsilon = \begin{cases} 1 - \frac{i}{500}, & \text{if } i \leq 450, \\ 0.1, & \text{if } 450 < i \leq 800, \\ 0.01, & \text{if } 800 < i \leq 1000, \\ 0, & \text{if } i > 1000, \end{cases} \quad (34)$$

where i is the current episode. The simulation parameters are summarized in Table I. The structure of the Q-network is shown in Table II, where the FC layer is an abbreviation of the full-connected layer. The compressed status matrix $\tilde{\mathbf{s}}_i^t$ is expanded into a one-dimensional tensor with length $L_{in} = D \times L$. The input tensor is divided into two branches after passing through the backbone network, one is the advantage function branch and the other one is the state value function branch, which are calculated as $A(\mathbf{s}, a)$ and $V(\mathbf{s})$, respectively. Finally, the Q-value function $Q(\mathbf{s}, a)$ is given by (31).

B. Performance Analysis

The global reward function obtained in the last time slot of each episode during the training period with $\beta = 0.5$ under different routing strategies is shown in Fig. 5. It can be seen that reward curves gradually rise and finally reach convergence

TABLE II
NETWORK STRUCTURE

Layer	Input Size	Output Size	Activation
BACKBONE			
Full-connected layer1	L_{in}	512	Relu
Full-connected layer2	512	512	Relu
Full-connected layer3	512	150	Relu
ADVANTAGE FUNCTION BRANCH			
Full-connected layer4	150	32	Relu
Full-connected layer5	32	16	Relu
Full-connected layer6	16	8	Linear
STATE VALUE FUNCTION BRANCH			
Full-connected layer7	150	32	Relu
Full-connected layer8	32	16	Relu
Full-connected layer9	16	1	Linear

under all routing strategies. The compromise in optimality is primarily caused by the partitioning of the holistic problem into independent decisions for each satellite, as well as the inaccuracies introduced by the division of global reward. The complexity of addressing the optimization issue for large-scale constellations holistically is prohibitively high, rendering some degree of loss inevitable for expeditious problem-solving. In terms of the latter issue, the effectiveness of reward decomposition is validated in Fig. 6 by comparing the sum of the

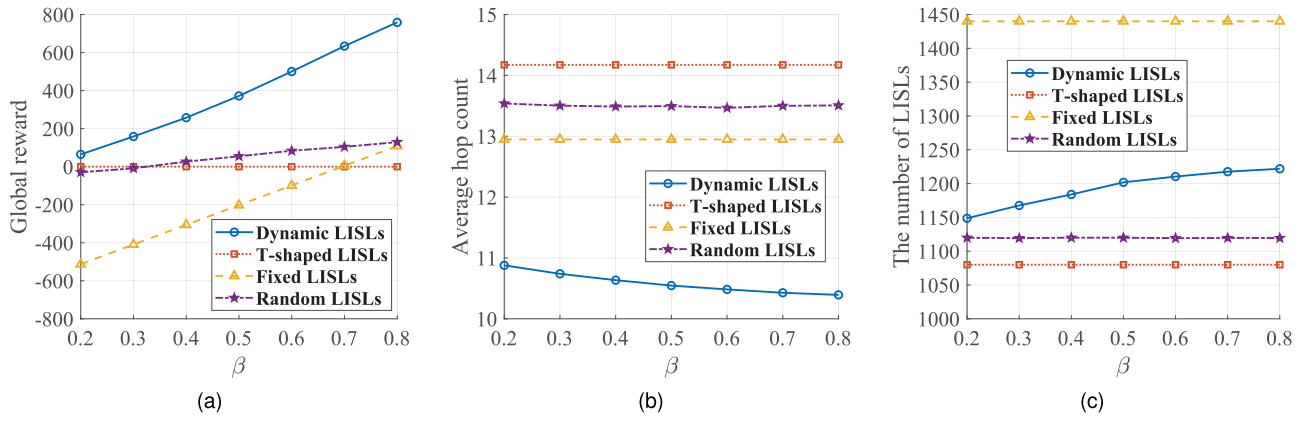


Fig. 7. Results with different β , strategy $\mathcal{R}_1$. (a) Global reward. (b) Average hop count. (c) The number of LISLs.

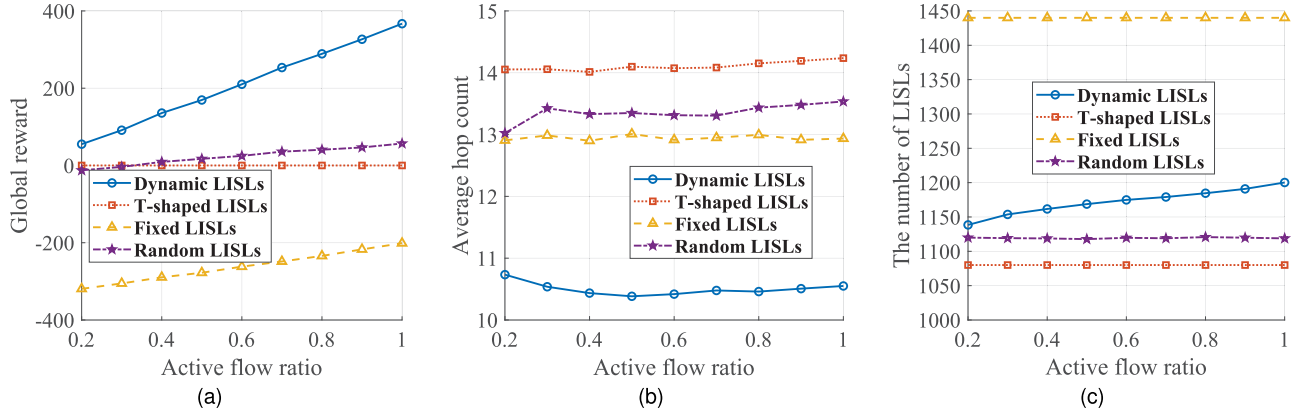


Fig. 8. Results with different active flows, strategy $\mathcal{R}_1$, $\beta = 0.5$. (a) Global reward. (b) Average hop count. (c) The number of LISLs.

agent rewards defined in (26) and global reward in (25). It is evident that the discrepancy between the sum of the agent rewards and global reward is not significant across all three routing strategies. The error is smallest under strategy $\mathcal{R}_1$, due to the majority of traffic through the dynamic LISLs being directed towards reducing hop count, rendering the analysis above (26) particularly pertinent. Conversely, the error is most pronounced under strategy $\mathcal{R}_2$, which solely considers the reduction of the maximum link load, thereby causing data flow to bypass a longer route to alleviate the load. The error under strategy $\mathcal{R}_3$ falls somewhere in between. Moreover, there is a strong correlation between the sum of agent rewards and global reward among the three strategies, with a correlation coefficient exceeding 0.98. Furthermore, other DRL methods utilizing global reward as a target struggle to achieve convergence in this scenario involving numerous agents. Consequently, the proposed method strikes a balance between the optimality loss and convergence.

The effectiveness of the proposed method is validated by contrasting it with other algorithms. (1) **Dynamic LISLs**, the proposed method. (2) **T-shaped LISLs**, consisting solely of the fixed T-shaped LISLs depicted in Fig. 2(b), without any dynamic LISLs. (3) **Fixed LISLs**, a configuration with four fixed LISLs as Fig. 2(a). (4) **Random LISLs**, consisting of T-shaped fixed LISLs and dynamic LISLs established randomly. The performance is compared across various scenarios.

1) *Results With Different β* : These methods were subjected to tests under varying β values, given the routing strategy $\mathcal{R}_1$. Global rewards defined in (25) are illustrated in Fig. 7(a). Average hop count and the number of LISLs of the four methods are shown in Fig. 7(b) and Fig. 7(c), respectively. Given that $(i, j) \in \mathcal{E}$ implies $(j, i) \in \mathcal{E}$, total LISLs count is calculated as $|\mathcal{E}|/2$. From global reward perspective, the proposed method consistently outperforms the others, regardless of the β value. T-shaped LISLs, serving as the benchmark defined in (25), maintain a constant global reward of zero. Conversely, due to the excessive number of established LISLs, fixed LISLs method, despite a reduction in its average hop count, only surpasses T-shaped LISLs in global reward when β is significantly large. This observation corroborates the assertion in [9] regarding the redundancy of the configuration with four fixed LISLs. The amount of LISLs and average hop count of random LISLs fall between those of T-shaped LISLs and fixed LISLs. Moreover, as β increases, the proposed method exhibits a gradual slowdown in the reduction of average hop count and increase in LISL count. This suggests a diminishing marginal utility in the addition of dynamic links for the reduction of average hop count.

2) *Performance in Different Active Flow Ratio*: During the model training, it is assumed that all data flows among nodes in $\mathcal{V}_0$ are active. However, traffic activity experiences fluctuations throughout the day in reality, with reduced traffic during rest

Algorithm 1 The MADRL for the Scheduling of Dynamic LISLs

Initialize:

- 1: Initialize routing policy $\mathcal{R}$.
- 2: Initialize the hot spot set $\mathcal{V}^h$ and the satellite network topology with $\mathcal{E}_d = \cdot$.
- 3: Initialize the current network and target network with random parameter, $\theta = \theta^-$.
- 4: Initialize the sensing matrix Φ_0 from a Gaussian distribution.
- 5: Initialize the global experience pool $\mathcal{Q}$ and the number of samples $N_e = 0$.
- 6: Initialize the number of episodes $EPISODE_NUM$.

Training:

- 7: **for** episode $k = 1$ to $EPISODE_NUM$ **do**
- 8: Generate the source and destination nodes set $\mathcal{V}_0$ from $\mathcal{V}^h$ and the traffic flow $\mathbf{F}^0$.
- 9: Execute the routing policy $\mathcal{R}$.
- 10: **for** slot $t = 0$ to $T - 2$ **do**
- 11: **for** agent $i = 1$ to N **do**
- 12: Each agent i obtains the environment state $\mathbf{s}_i^t$.
- 13: Compress the state information $\tilde{\mathbf{s}}_i^t \leftarrow \Phi_0 \mathbf{s}_i^t$.
- 14: Input the $\tilde{\mathbf{s}}_i^t$ to the current network, and select the action a_i^t with the ϵ -greedy strategy according to (27).
- 15: Execute the action a_i^t .
- 16: **end for**
- 17: Generate the traffic flow $\mathbf{F}^{t+1}$ and execute the routing policy $\mathcal{R}$.
- 18: **for** agent $i = 1$ to N **do**
- 19: Calculate the reward r_i^t according to (26).
- 20: Obtain the state in next slot $\mathbf{s}_i^{t+1}$ and $\tilde{\mathbf{s}}_i^{t+1} \leftarrow \Phi_0 \mathbf{s}_i^{t+1}$.
- 21: Store the experience sample $\langle \tilde{\mathbf{s}}_i^t, a_i^t, r_i^t, \tilde{\mathbf{s}}_i^{t+1} \rangle$ into global experience pool $\mathcal{D}$ and update N_e .
- 22: **end for**
- 23: **if** $N_e > N_s$ **then**
- 24: Sample the experience $\langle \tilde{\mathbf{s}}_j^t, a_j^t, r_j^t, \tilde{\mathbf{s}}_j^{t+1} \rangle$ from $\mathcal{D}$ according to (33).
- 25: Calculate the loss function by (30).
- 26: Update the parameter of the current network θ and share with all the agents.
- 27: **end if**
- 28: **end for**
- 29: Update the parameter of target network with $\theta^- \leftarrow \theta$ each C slots.
- 30: **end for**

periods and increased data during work periods. Consequently, tests are predicated on the assumption that only a certain percentage of data flows remain active. Test results of the four methods under routing policy $\mathcal{R}_1$ with $\beta = 0.5$ are presented in Fig. 8. Results suggest that with the rise in the proportion of active data flows, the number of LISLs in the proposed method incrementally ascends, while the average hop count initially descends and then slightly ascends. This is attributable to the fact that an increase in the number of links

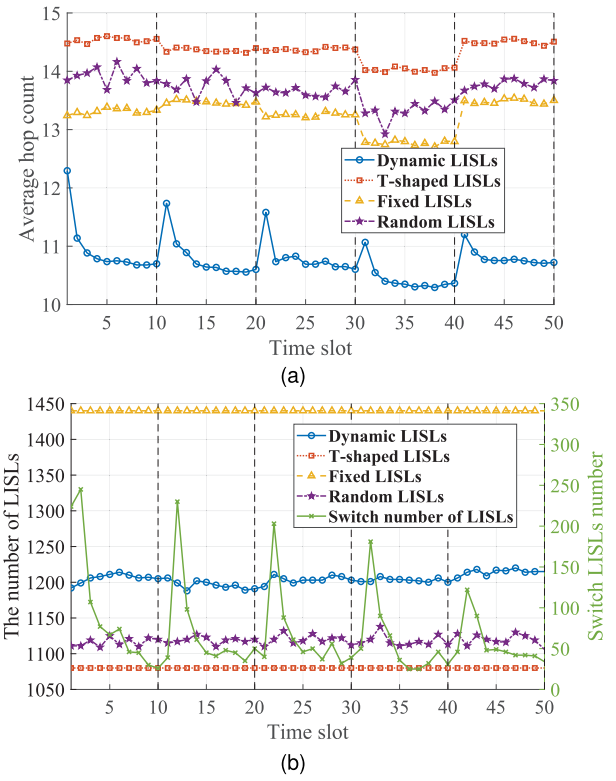


Fig. 9. The performance over multiple slots, $\beta = 0.5$, strategy $\mathcal{R}_1$. (a) Average hop count. (b) The number of LISLs.

can effectively reduce the average hop count when the load is low. However, as the traffic load intensifies further, some links become fully loaded, leading to a corresponding increase in average hop count. Despite this, the overall performance of the proposed method remains stable and significantly outperforms other three methods.

3) *Performance Over Multiple Slots:* Due to the continuous movement of satellites, access points $\mathcal{V}_0$ are also changing among different episodes. Performance under routing strategy $\mathcal{R}_1$ with $\beta = 0.5$ is tested over 50 continuous time slots, i.e., five episodes. Results are shown in Fig. 9, with each episode divided by a dashed line. Average hop count for the four methods is shown in Fig. 9(a). At the first slot of each episode, average hop count of the proposed method increases significantly because the link scheduling is still based on the traffic load of the previous episode. However, it still outperforms other methods and can quickly reach a steady state. The number of LISLs of the four methods and the switch number of dynamic LISLs are shown in Fig. 9(b). It can be seen that the link switching will peak at the second time slot of each episode because the scheduling in this slot is the first based on the current episode information, and then the number of link switches rapidly decreases. In summary, the proposed method can quickly adapt to the changes in traffic access points caused by satellite movement.

4) *Results Under Different Routing Strategy:* In order to verify the effectiveness of the proposed method under different routing strategies, a simulation is performed among 50 episodes under three routing strategies with $\beta = 0.5$. The network snapshots at the end of each episode are shown in Fig. 10, and the numerical results are listed in Table III.

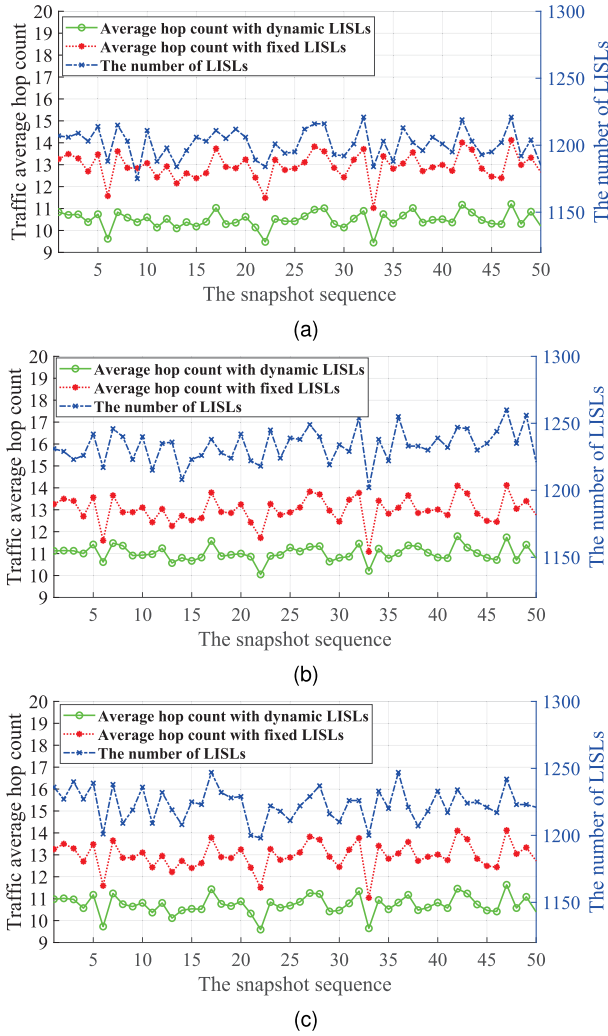


Fig. 10. The snapshot of the satellite network, $\beta = 0.5$. (a) Strategy $\mathcal{R}_1$. (b) Strategy $\mathcal{R}_2$. (c) Strategy $\mathcal{R}_3$.

TABLE III
PERFORMANCE COMPARISON BETWEEN FIXED
LISLs AND DYNAMIC LISLs

Routing Strategy	$\mathcal{R}_1$	$\mathcal{R}_2$	$\mathcal{R}_3$
Hop count with fixed LISLs	12.94	13.00	12.97
Hop count with dynamic LISLs	10.52	11.01	10.73
Hop count reduction	2.42	1.99	2.24
LISLs number with fixed LISLs	1440	1440	1440
LISLs number with dynamic LISLs	1200.86	1233.2	1223.3
LISLs number reduction(%)	16.62	14.36	15.05

The results demonstrate that the proposed method can effectively reduce the average hop count and the number of LISLs under various strategies. The average delay can be reduced by about two hops, and a strong correlation exists between the average hop count with dynamic LISLs and that of fixed LISLs. Moreover, the average hop count gap between three routing strategies is increased after dynamic link scheduling, which is attributed to the detour of data flows under strategy $\mathcal{R}_2$ and $\mathcal{R}_3$. In terms of the amount of LISLs, the proposed method reduces the amount of LISLs under all

three routing strategies by about 15% compared with fixed LISLs. In addition, it can be observed that the number of established LISLs varies significantly under different routing policies. Among three routing strategies, $\mathcal{R}_1$, which focuses solely on the average hop count, has the fewest dynamic LISLs. $\mathcal{R}_2$, which focuses exclusively on link load, has the most dynamic links, and $\mathcal{R}_3$ lies somewhere in between the two. The reason is that when only the average hop count is considered, the delay can be reduced by establishing several critical paths. On the other hand, when the link load is taken into account, the load must be distributed by establishing more links.

C. Implementation and Complexity Analysis

1) *Practical Implementation*: During the training process of DRL, frequent data transmission is required. The delay requirement for transmission of LEO satellite state information and DNN training is not stringent, since these processes can be executed concurrently with the LISL scheduling. Therefore, a practical implementation is to utilize a GEO satellite as data storage and training center, while each LEO satellite makes decisions with its current parameters. Two additional GEO satellites are required to act as relays and collect information from LEO satellites within their coverage areas due to the limited coverage of single GEO satellite. The data center GEO satellite receives information directly from the LEO satellites within its coverage area, as well as relayed information from the other two GEO satellites, thereby forming a global experience pool, and updating the DNN parameters accordingly. The updated model parameters are broadcasted to LEO satellites in a similar manner. The longest path from the LEO satellite to the data center GEO satellite requires passing through a relay GEO satellite with a maximum round-trip time of 629 ms. This is adequate to complete the reception of LEO satellite state information and model parameter updates within a time slot.

2) *Computational Complexity*: In the MADRL algorithm, most computations are associated with the calculation of DNNs. The computational complexity of the FC layer directly correlates with its parameter count, which can be calculated as $\sum_{i=1}^{L_{num}} (L_i^{in} \times L_i^{out} + L_i^{out})$, where L_{num} denotes the number of layers in DNN, and L_i^{in} and L_i^{out} represent the input and output size of the i -th layer, respectively. The computational complexity can be estimated as $\mathcal{O}(\sum_{i=1}^{L_{num}} L_i^{in} \times L_i^{out})$. Therefore, utilizing compressed samples as inputs reduces the computational overhead to 20% of using original samples as inputs, and also requires less memory. Additionally, the DNN adopted in this paper necessitates fewer than 10^6 floating-point multiplications, which can be calculated within 1 ms with the specialized hardware.

3) *Communication and Storage Overhead*: During time slot $t + 1$, satellite $i \in \mathcal{V}$ uploads information from the previous slot, including state $\mathbf{s}_i^t$ (10×81 float32 matrix), action a_i^t (one-byte value), and reward r_i^t (float32 number), resulting in a transmission of 3,245 B per time slot. Each relay GEO satellite is responsible for relaying data from $N_{relay} = 240$ LEO satellites it covers, leading to relay 760.55 KB per time slot. For the experience pool on the data center GEO satellite, each

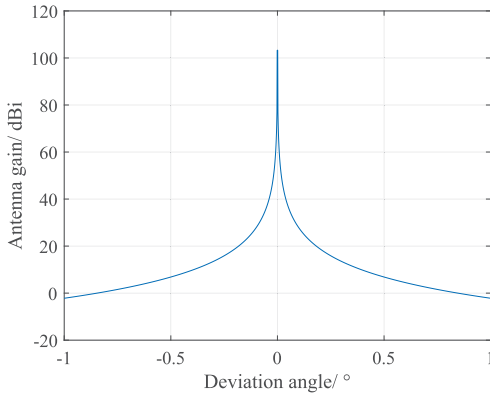


Fig. 11. The antenna gain pattern introduced in [42] and [43].

experience sample is formatted as $\langle \mathbf{s}_i^t, a_i^t, r_i^t, \mathbf{s}_i^{t+1} \rangle$, requiring 6485 B. With not more than 10^6 samples, it necessitates storage of 6.04 GB. The DNN parameters are saved in float16 format, and thus GEO satellites need to broadcast 1.46 MB when the model is updated. Therefore, the communication cost during the training of DNN for the satellite network is acceptable.

VI. CONCLUSION

In this paper, a dynamic LISLs scheduling algorithm combined with a specific route strategy is proposed. A link mode consists of three fixed LISLs and one LISL is investigated and verified, which fully realizes the potential of long-distance communication by space-borne laser terminal. The optimization objective is constructed by a tradeoff between establishing dynamic links to reduce hops and shutting down idle laser terminals to minimize the energy consumption associated with the closed-loop control and then transformed into an MDP. The complexity of the holistic optimization problem is significantly reduced by decomposing it into independent decisions for each satellite. The convergence of the algorithm is ensured by introducing DNN, prioritized experience replay, and sharing of network parameters and experience pool. Meanwhile, the CS is used to compress the link state information, which effectively reduces the communication, storage, and computing overhead. Finally, the simulation results show that the proposed algorithm can significantly decrease the average delay by about two hops and energy consumption of satellite networks by around 15% under various routing strategies after offline training without additional hardware costs.

APPENDIX A

CALCULATION OF SPATIAL PARAMETERS

The laser terminal parameters in [4] and the antenna gain described in [42] and [43], as shown in Fig. 11, are employed to validate the feasibility of establishing dynamic LISLs and to evaluate potential interference with the fixed LISLs. Therefore, the spatial parameters of dynamic LISLs necessitate to be calculated first.

As shown in Fig. 12, the angle between the two OPs is Ω , and the inclination angle is δ . The phase angle of the satellite's current position in OP, which can be calculated with the

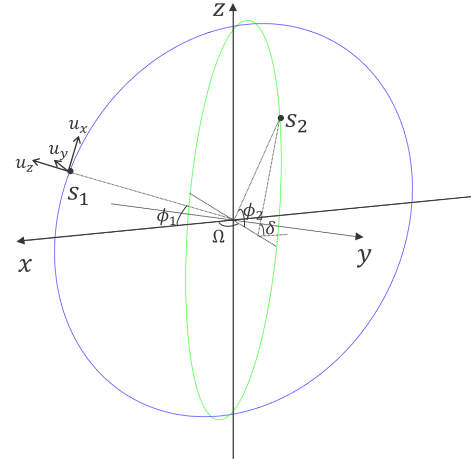


Fig. 12. The schematic diagram of the geometric relationship between two satellites.

ascending node of the OP as the starting point, is represented by ϕ . For simplicity, the line connecting the ascending node and the descending node of one of the OPs is taken as the x-axis. The vector position of S_1 is calculated as

$$\mathbf{S}_1 = R \begin{pmatrix} \cos(\phi_1) \\ \sin(\phi_1) \cos(\delta) \\ \sin(\phi_1) \sin(\delta) \end{pmatrix}, \quad (35)$$

and that of S_2 is given by

$$\mathbf{S}_2 = R \begin{pmatrix} \cos(\phi_2) \cos(\Omega) - \sin(\phi_2) \cos(\delta) \sin(\Omega) \\ \cos(\phi_2) \sin(\Omega) + \sin(\phi_1) \cos(\delta) \cos(\Omega) \\ \sin(\phi_2) \sin(\delta) \end{pmatrix}, \quad (36)$$

where ϕ_1 and ϕ_2 indicate the orbit phase angle of S_1 and S_2 , respectively, and $R = 7,587$ km indicates the radius of the orbit. Therefore, the pointing vector from S_1 to S_2 can be represented as $\mathbf{S}_2 - \mathbf{S}_1$, and the communication distance can be calculated as $\|\mathbf{S}_2 - \mathbf{S}_1\|_2$. To obtain the azimuth and elevation angles of the terminal, a coordinate is established with S_1 as the reference frame, and the unit vectors in three orthogonal directions are given by

$$\begin{cases} \mathbf{u}_x = (-\sin(\phi_1), \cos(\phi_1) \cos(\delta), \cos(\phi_1) \sin(\delta))^T, \\ \mathbf{u}_y = (0, -\sin(\delta), \cos(\delta))^T, \\ \mathbf{u}_z = (\cos(\phi_1), \sin(\phi_1) \cos(\delta), \sin(\phi_1) \sin(\delta))^T. \end{cases} \quad (37)$$

Therefore, the elevation angle ϵ can be calculated as

$$\epsilon = \tan^{-1} \left(\frac{(\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_z}{\sqrt{((\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_x)^2 + ((\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_y)^2}} \right), \quad (38)$$

and the azimuth angle α can be obtained as

$$\alpha = \begin{cases} \tan^{-1} \left(\frac{(\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_y}{(\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_x} \right), & \text{if } (\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_x > 0, \\ \tan^{-1} \left(\frac{(\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_y}{(\mathbf{S}_2 - \mathbf{S}_1)^T \mathbf{u}_x} \right) + \pi, & \text{others.} \end{cases} \quad (39)$$

The communication distance does not exceed 4,000 km according to Fig. 13(a), while the beam pointing range is

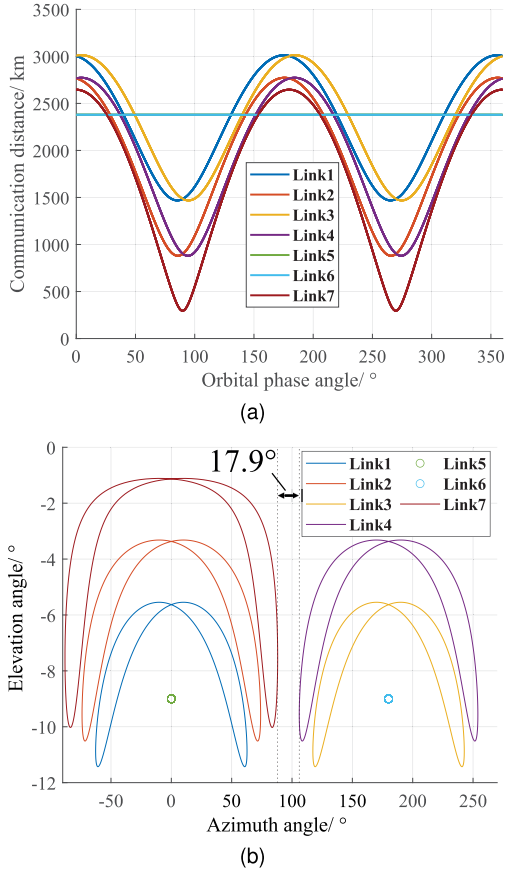


Fig. 13. The results for distance and angle range. (a) Communication distance. (b) Elevation angle and azimuth angle.

shown in Fig. 13(b), the range of elevation angle is between -12° and 0° , within the steering range of the terminal in [4]. For the azimuth angle, there is a gap of 17.90° between the azimuth range of each link, which can meet the limits of the terminal, whose pointing blind zone range is 10° .

Since the altitudes of the OPs in the constellation are the same, the satellites have identical angular velocities and can be given by

$$\omega = \sqrt{\frac{GM}{R^3}}, \quad (40)$$

where G is the gravitational constant and M represents the mass of the Earth. Thus, the difference $\Delta\phi$ between ϕ_1 and ϕ_2 is constant, therefore, after substituting $\phi_2 = \phi_1 + \Delta\phi$ into (38) and (39), ϕ_1 is the only time-varying factor in ϵ and α . As a result, the slew rate of the azimuth angle and elevation angle can be calculated as

$$\begin{cases} \omega_\epsilon = \frac{d\epsilon}{d\phi} \frac{d\phi}{dt} = \frac{d\epsilon}{d\phi} \omega, \\ \omega_\alpha = \frac{d\alpha}{d\phi} \frac{d\phi}{dt} = \frac{d\alpha}{d\phi} \omega, \end{cases} \quad (41)$$

where ω_ϵ and ω_α represent the slew rate of elevation angle and azimuth angle, respectively. The results indicate that the steering rate can also meet the requirements, as shown in Fig. 14.

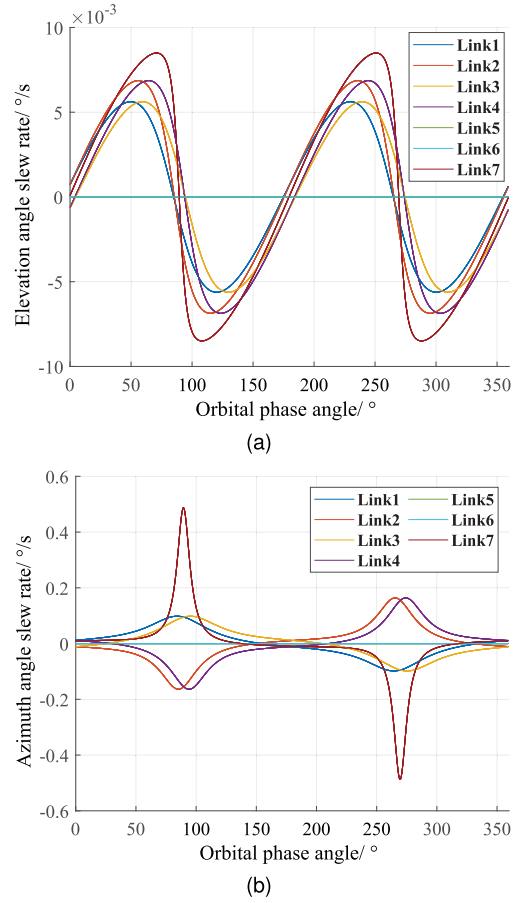


Fig. 14. The results for the angle steering rate. (a) Elevation angle. (b) Azimuth angle.

APPENDIX B ANALYSIS OF INTERFERENCE BETWEEN DYNAMIC LISLS AND FIXED LISLS

Given the close direction of the laser beam, the potential interference between dynamic intra-OP LISLS and fixed intra-OP LISLS should be considered. As determined in [43], the signal-to-interference ratio, in this case, is primarily dependent on the difference of the path loss, and the transmit/receive antenna gain. The distance between the two receivers is negligible compared to the transmission distance of the laser beams, and so the transmit antennas can be assumed to be aligned. Regarding the receiving terminal, the pointing vectors of fixed LISLS can be calculated with the same method as mentioned above. Subsequently, the angular difference between the fixed LISLS and dynamic LISLS at the receiver can be obtained by the inner product. Assuming a perfect alignment of the beams, the calculated minimum angular separation between fixed LISLS and dynamic LISLS is 1.13° . As shown in Fig. 11, the antenna gain at the receiving end in this scenario is below 0 dBi, significantly lower than the maximum gain of 104 dBi with beam alignment, which is attributed to the extremely narrow laser beam width of tens of microradians.

Moreover, compared to the substantial reduction in receive gain, the difference in path loss has a negligible effect on received power. Thus, the interference from dynamic LISLS to fixed LISLS can be considered negligible based on the preceding discussions.

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